Sr. Software Systems Engineer - Cardiac Ablation Solutions

Medtronic · Mounds View, MN; Newton, MA

Job ID (this archive)	HI-145
Employer	Medtronic
Location(s)	Mounds View, MN; Newton, MA
Career stage (archive tag)	Experienced
Location group	Other U.S.
Majors (archive tags)	Systems Engineering
Pay listed	$120,800 – $181,200 per year
Employment type	Full time
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Employer requisition	R74266
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Source page	https://medtronic.wd1.myworkdayjobs.com/en-US/MedtronicCareers/job/Mounds-View-Minnesota-United-States-of-America/Sr-Software-Systems-Engineer_R74266-1
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Job description

We anticipate the application window for this opening will close on - 12 Oct 2026

Careers that change lives start here. Medtronic is a global leader in healthcare technology with a Mission to alleviate pain, restore health, and extend life. Our 95,000 employees work across more than 150 countries to put patients first - developing innovative medical technologies that improve the lives of 72+ million patients each year. Your unique talents will help shape the future of healthcare while building a career grounded in purpose, growth, and impact.

A Day in the LifeAt Medtronic, we bring bold ideas forward with speed and decisiveness to put patients first in everything we do. In-person exchanges are invaluable to our work. We’re working a minimum of 4 days a week onsite as part of our commitment to fostering a culture of professional growth and cross-functional collaboration as we work together to engineer the extraordinary.

Medtronic is seeking a motivated and detail-oriented Sr. Software Systems Engineer to join our Affera Mapping and Ablation System team, supporting the design and development of advanced software-enabled capital systems for the cardiac electrophysiology lab.

The Sr. Software Systems Engineer will an integral member in the Affera Software Team, working on cutting-edge technology in the electrophysiology field. This role offers a rare chance to be at the forefront of helping develop a groundbreaking technology that can disrupt the industry.

The ideal candidate has strong MBSE skills and experience using SysML (v1/v2) to support V-model traceability from requirements through verification. They will own key features, collaborate across engineering teams, and are enthusiastic about making a meaningful difference for millions.

About the Affera Mapping and Ablation System

The Affera Mapping and Ablation System is an innovative medical technology designed to treat patients with cardiac arrhythmias. The system involves advanced mapping techniques to create detailed images of the heart's electrical activity, helping physicians identify abnormal areas causing arrhythmias. Once identified, the Affera Ablation System delivers targeted

energy to disrupt these abnormal signals, restoring

Primary Responsibilities

Develop and maintain system and software architecture models using SysML (v1/v2).

Apply MBSE principles to connect system intent to software services, APIs, schemas, modules, and firmware.

Define interface contracts and allocations between system functions, software, and hardware components.

Support V-model traceability across requirements, design, implementation, and verification

Author and review software system requirements with full bidirectional traceability.

Perform risk analysis activities including FMEA and interface risk identification.

Contribute to process and tooling improvements for MBSE adoption and trace automation.

Apply the appropriate standards, processes, procedures, and tools throughout the system development life cycle.

Required Qualifications

Bachelor's degree and a minimum of 4 years of relevant experience

OR Master’s degree with a minimum of 2 years of relevant experience

OR PhD with 0 years relevant experience

Experience with requirements-driven development and traceability

Familiarity with MBSE concepts and SysML

Strong understanding of good documentation practices

Preferred Qualifications

Knowledge of Medical Product development, IEC-62304, ISO 13485, and FDA guidance in a technical discipline, preferably software related

4+ years of software verification experience

Proficient in programming languages such as Python, C++/C# or similar languages

Experience with SysML v2 tooling.

Experience writing technical reports

Experience with QT/QML

Experience with software version control systems, such as Git

Experience with Agile development

Strong work ethic and entrepreneurial drive

Ability to work effectively in a fast-paced, collaborative environment.

Strong communication skills, both written and verbal

#LI-MDT

For Baccalaureate degrees earned outside of the United States, a degree that satisfies the requirements of 8 C.F.R. § 214.2(h)(4)(iii)(A) is required.

Physical Job Requirements

The above statements are intended to describe the general nature and level of work being performed by employees assigned to this position, but they are not an exhaustive list of all the required responsibilities and skills of this position.

The physical demands described within the Responsibilities section of this job description are representative of those that must be met by an employee to successfully perform the essential functions of this job. Reasonable accommodations may be made to enable individuals with disabilities to perform the essential functions. For Office Roles: While performing the duties of this job, the employee is regularly required to be independently mobile. The employee is also required to interact with a computer, and communicate with peers and co-workers. Contact your manager or local HR to understand the Work Conditions and Physical requirements that may be specific to each role.

U.S. Work Authorization & Sponsorship

At Medtronic, we are committed to fostering an environment where employees can thrive and make a meaningful impact. In alignment with our enterprise-wide workforce planning approach, U.S. work authorization sponsorship (H-1B, TN, J, etc.) is offered exclusively for Principal-level roles and above, where specialized expertise aligns with long-term business needs. Roles below the Principal level require candidates to possess unrestricted U.S. work authorization at the time of hire and for the

duration of employment.

Recruitment Fraud Alert

We are aware of phishing scams targeting job seekers. Please keep the following in mind:

Apply only through official Medtronic channels. All legitimate Medtronic recruiting communications come from approved Medtronic platforms and official @medtronic.com email addresses.

Medtronic will never ask for payment or sensitive personal information (such as bank account or Social Security details) during early stages of the hiring process. Any such requests are not legitimate.

If you receive a suspicious message claiming to be from Medtronic, do not respond, click links, or open attachments.

If you have any questions, concerns regarding the authenticity of a communication alleged to have been made by or on behalf of Medtronic, please contact us immediately at AskHR@medtronic.com.

Benefits & Compensation

Medtronic offers a competitive Salary and flexible Benefits Package

A commitment to our employees lives at the core of our values. We recognize their contributions. They share in the success they help to create. We offer a wide range of benefits, resources, and competitive compensation plans designed to support you at every career and life stage.

Salary ranges for U.S (excl. PR) locations (USD):$120,800.00 - $181,200.00

This position is eligible for a short-term incentive called the Medtronic Incentive Plan (MIP).

The base salary range is applicable across the United States, excluding Puerto Rico and specific locations in California. The offered rate complies with federal and local regulations and may vary based on factors such as experience, certification/education, market conditions, and location. Compensation and benefits information pertains solely to candidates hired within the United States (local market compensation and benefits will apply for others).

The following benefits and additional compensation are available to those regular employees who work 20+ hours per week: Health, Dental and vision insurance, Health Savings Account, Healthcare Flexible Spending Account, Life insurance, Long-term disability leave, Dependent daycare spending account, Tuition assistance/reimbursement, and Simple Steps (global well-being program).

The following benefits and additional compensation are available to all regular employees: Incentive plans, 401(k) plan plus employer contribution and match, Short-term disability, Paid time off, Paid holidays, Employee Stock Purchase Plan, Employee Assistance Program, Non-qualified Retirement Plan Supplement (subject to IRS earning minimums), and Capital Accumulation Plan (available to Vice Presidents and above, or subject to IRS earning minimums).

Regular employees are those who are not temporary, such as interns. Temporary employees are eligible for paid sick time, as required under applicable state law, and the Employee Stock Purchase Plan. Please note some of the above benefits may not apply to workers in Puerto Rico.

Further details are available at the link below:

Medtronic benefits and compensation plans

It is the policy of Medtronic to provide equal employment opportunity (EEO) to all persons regardless of age, color, national origin, citizenship status, physical or mental disability, race, religion, creed, gender, sex, sexual orientation, gender identity and/or expression, genetic information, marital status, status with regard to public assistance, veteran status, or any other characteristic protected by federal, state or local law. In addition, Medtronic will provide reasonable accommodations for qualified individuals with disabilities.

If you are applying to perform work for Medtronic, Inc. (“Medtronic”) in any position which will involve performing at least two (2) hours of work on average each week within the unincorporated areas of Los Angeles County, you can find here a list of all material job duties of the specific job position which Medtronic reasonably believes that criminal history may have a direct, adverse and negative relationship potentially resulting in the withdrawal of a conditional offer of employment. Medtronic will consider for employment qualified job applicants with arrest or conviction records in accordance with the Los Angeles County Fair Chance Ordinance for Employers and the California Fair Chance Act.

Notes for ECS 1100 students

Course spreadsheet notes

Skills mentioned: SysML; MBSE; V-model traceability; FMEA; software/hardware interfaces.

Systems Engineering

Direct role in this area. See employer’s specific degree disciplines.

Degree: Bachelor’s + 4 years OR master’s + 2 years OR PhD + 0 years

Experience / eligibility: Requirements-driven development, traceability, MBSE and SysML familiarity required.

Other: No work-visa sponsorship. See full posting for local conditions. Posted closing date: 2026-10-12.

Other majors named: Degree discipline not specified

Career stage: Experienced / senior / Full-time / experienced

Salary evidence: Salary ranges for U.S (excl. PR) locations (USD):$120,800.00 - $181,200.00